

NHLANHLA MOKOENA

AI Engineer | Agentic AI, RAG & MLOps

Johannesburg, South Africa

- LinkedIn Profile [Link](#)
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Production-focused AI Engineer specializing in agentic workflows, RAG architectures, and end-to-end MLOps using LangChain, LangGraph, and N8N. Proven expertise in building robust Python backends, Dockerized deployments, CI/CD pipelines, and translating mathematical concepts into production-grade systems. Currently engineering XR simulation platforms and automated model evaluation frameworks at Nudle.

CORE TECHNICAL SKILLS

- **Agentic AI & GenAI:** LangChain, LangGraph, n8n, RAG Pipelines, Vector Databases, Multi-Agent Workflows, Claude & OpenAI APIs
- **Frameworks & Languages:** Python, FastAPI, Next.js, React, Node.js
- **ML & Deep Learning:** PyTorch, Scikit-Learn, Weights & Biases, Model Training, Fine-Tuning, Inference & Evaluation
- **Data & Libraries:** Pandas, NumPy, Matplotlib, SQL, ETL Pipelines
- **MLOps & Delivery:** Docker, Git, GitHub Actions (CI/CD), End-to-End MLOps Pipelines, Serverless Deployments
- **Cloud & Tools:** AWS, Microsoft Azure, Hugging Face Spaces, Render, Vercel
- **Architecture & Systems:** System Design, Scalable Architecture, Microservices Design

EXPERIENCE

AI Engineer

Nudle | XR Simulations | May 2025 – Present

- **Developed a headless Blender 3D AI agent framework** that executes programmatic Python scripts to automatically build, bake, and render structural 3D lesson environments based on dynamic curricular inputs.
- **Architected multi-modal GenAI pipelines** integrating **Trellis** for automated, high-fidelity 3D asset generation directly from text/image prompts, slashing manual 3D modeling overhead by 65%.
- **Orchestrated text-to-video workflows** using the **Wan2.2 video generation model** to render cinematic, domain-specific educational lesson animations with ultra-low generation latency.
- Scaled **Docker-containerised GPU** serving to deliver sub-second cold starts and maximum throughput for production video, image, and 3D generation AI models.

AI Trainer

Alignerr | AI Model Training | Dec 2024 – Jan 2025

- **Improved dialogue safety and reasoning alignment**, optimizing model adherence across edge-case benchmarks, by systematically scoring, red-teaming, and refining model responses.
- **Conducted deep-dive alignment testing** on LLMs, identifying behavioral drift and improving conversational reasoning.
- **Evaluated complex model outputs** to enhance dialogue safety, ensuring strict compliance with safety guidelines and behavioral guardrails.

Junior AI Software Developer

Artintel | AI Software | Oct 2023 – Sep 2024

- **Converted** unstructured data into clean tokens for **LLM training / RAG applications**.
- **Integrated Hugging Face** transformers and **PyTorch** frameworks into a continuous integration (**CI/CD**) pipeline for automated model evaluation.
- Lowered computational overhead and abstracting model complexities via **REST APIs** for non-technical users.

PROJECTS

Avatar-3D Pipeline | [Link](#)

- Engineered a "Director & Marionette" text-to-animation engine, translating natural language into 24-dimensional kinematic skeletal animations with strict schema compliance, by using **Llama-3 (via Groq API)** with strict motion validation.
- Scaled interactive WebXR content delivery, supporting multi-user high-concurrency requests, by containerizing the Python backend for **Hugging Face Spaces** and building a **Next.js/React Three Fiber** frontend.

Orbit-3D Asset Pipeline | [Link](#)

- Automated high-fidelity 3D model generation, eliminating manual asset texturing and decimation bottlenecks, by building a multimodal pipeline integrating Tripo3D.
- Streamlined asset optimization for WebXR/Three.js, accelerating mesh rendering speeds, by containerizing headless Blender processing engines into Docker microservices.
- Reduced asset-generation job processing latency, preventing backend timeouts during bulk generation, by architecting an asynchronous, event-driven task system with FastAPI and Redis queues.

Real Estate Price Predictor | [Link](#)

- Built a high-accuracy property valuation pipeline, achieving an R^2 score of 0.9037 and RMSE of 1,141, by engineering an end-to-end XGBoost machine learning regression model in Python.
- Building and wrapped the model with FastAPI REST service, containerized with Docker and automated via GitHub Actions CI/CD.

EDUCATION

University of Witwatersrand | 2015 -2019

- Bsc in Biological science
Incomplete
- Bsc in Mechanical Engineering
Incomplete

ALX/ExploreAI Academy | Jun 2023 - Sep 2024

- Data Science

DynamicDNA ICT Academy | May 2023 - Aug 2024

- Systems Development
NQF Level 4

CERTIFICATIONS

- **DataCamp** Associate AI Engineer for Developers [Link](#)
- **Microsoft** Azure Fundamentals (AZ-900) [Link](#)
- **Microsoft** 365 Certified [Link](#)
- **Udacity** Introducing Generative AI with AWS [Link](#)
- **ALX/ExploreAI** AI Career Essentials [Link](#)
- **365 Data Science** Credit Risk Modelling in Python & Machine Learning [Link](#)

